CMINet: A Unified Cross-Modal Integration Framework for Crop Classification From Satellite Image Time Series

Yutong Hu, Qingwu Hu, and Jiayuan Li

***Abstract*— Accurate automatic interpretation of crops is crucial for agricultural monitoring and food security assessment. In recent years, there has been a rise in platforms generating multimodal and multitemporal remote sensing images at an unprecedented speed. These images provide rich temporal, spatial, and spectral information, enabling more comprehensive land cover classification. Therefore, single-modal crop classification can benefit from complementary modalities. However, given the distinct characteristics of different modality sensors, enhancing the performance of deep networks by integrating diverse modality data remains a significant challenge. Unlike previous methods, this article proposes a two-stage fusion network named CMINet for crop classification from satellite image time series (SITS). Specifically, we adopt a decoupled framework to encode the spatial and temporal features of SITS. At each level of the spatial encoder, we design a cross-modal transfer module (CMTM) to complement bimodality branch features by transferring knowledge from one modality to rectify the features of another modality. After rectified feature pairs are encoded by a temporal encoder, we develop a cross-attention fusion module (CAFM) to conduct adequate context exchange before merging. The seamless combination of these novel designs forms a robust multimodal representation, outperforming the state-of-the-art methods on two public multimodal crop classification datasets. Compared to existing methods, our CMINet improves at least 1.1% OA, 1.7% mF1, and 2.0% mIoU on the PASITS-R dataset and 2.6% OA, 1.1% mF1, and 2.8% mIoU on the South Sudan dataset.**

***Index Terms*— Crop classification, cross-modal fusion, deep learning, satellite image time series (SITS).**

I. Introduction

With the advancement of remote sensing technology, an increasing amount of remote sensing image data from different modalities such as Landsat, MODIS, Sentinel-1, and Sentinel-2 is available. These data sources contain abundant spatial and spectral information that can greatly aid in classification tasks. By incorporating multiple modalities in classification models, it is workable to utilize complementary information and overcome the limitations of single-modal data classification models [1]. For instance, optical remote sensing images are frequently compromised by cloud cover, leading to partial data loss. Employing synthetic aperture radar (SAR) images, which can penetrate clouds, as an auxiliary data source can effectively mitigate this issue [2]. Fusion of data from different modalities can be achieved by training a single neural network with multimodal input [3] or by using multiple networks to process data from each modality and then integrating the results [4]. Multimodal classification enables us to fully exploit the advantages of diverse data sources, minimize the influence of atmospheric interference such as clouds and fog, and consequently enhance the accuracy and reliability of classification. In recent years, the field of machine learning and data mining has witnessed the wide application of deep learning, which has proven to be a groundbreaking technology for various tasks including object detection and image classification [6], [7]. Nevertheless, current deep networks for crop classification mainly focus on utilizing single-modal optical image data. This approach poses limitations as optical sensors are prone to challenges such as cloud cover and atmospheric refraction. Consequently, incomplete information extraction and susceptibility to cloud and atmospheric interference negatively impact the accuracy of classification [8]. In contrast, radar images possess the advantage of high revisit frequency and insensitivity to cloud cover, making them an ideal choice for monitoring rapid biological changes in agricultural areas [9]. By simultaneously leveraging the strengths of both optical and radar images, a crop classification approach can enable the extraction of feature information from both modalities, thereby mitigating the impact of clouds and atmosphere [10].

Using only single-modal remote sensing images for crop classification can already achieve high accuracy [11], [12]. However, significant improvements in accuracy and precision can be attained when combining Sentinel-2 data with other modalities [13], [14]. Among various modality combinations discussed in studies on multimodal fusion, the fusion of optical images and SAR images is the most commonly explored. Research has demonstrated that a winter land use classification model, utilizing a combination of Sentinel-2 and Sentinel-1 images, exhibited higher overall accuracy compared to models using only Sentinel-2 or Sentinel-1 images [15]. Furthermore, terrain information can be considered as an additional modality to enhance the distinction of land cover [16]. Apart from multimodal data combinations, it is essential to consider fusion strategies for multimodal data. Earlier studies on multimodal

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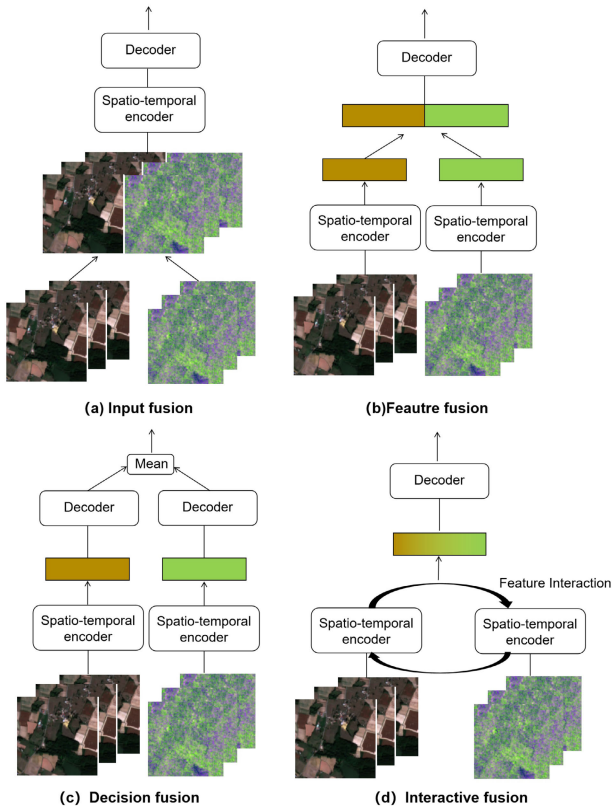


Fig. 1. **Fusion strategies.** (a) Raw SITS are concatenated at the input layer of the network. (b) Features learned from each modality are concatenated before the decoder. (c) Each modality branch conducts independently, and results are averaged at the decision level. (d) Each modality branch performs sufficient information transmission within the network, resulting in complementary features.

data fusion mainly employed direct channelwise concatenation of features from various modalities, without considering whether these features underwent information transfer and rectification for integration [17], [18]. Fig. 1(a)–(c) shows the three common fusion strategies currently. However, this approach has been shown to suppress representative features within and between modalities [19], [20]. To address this, a more interactive fusion strategy, as shown in Fig. 1(d), for multimodal data should enable fusion at multiple layers of a deep model.

Based on the above considerations, in this article, we propose CMINet, a deep interactive cross-modal fusion network for crop classification from satellite image time series (SITS). Specifically, CMINet is constructed as a dual-branch architecture, with a spatial encoder and a temporal encoder adapted to separately model spatial and temporal features from the two modalities of the SITS. Two specific modules are designed for feature transfer and feature fusion between modalities.

1) The cross-modal transfer module (CMTM) integrates the bimodality features by facilitating channelwise and spatialwise transfer. This mechanism transmits useful knowledge from one modality to supplement the features of another, enabling both branches to concentrate more on the complementary information from each other. In addition, it mitigates the effects of uncertainties and noise measurements from different modalities, thereby enhancing the overall feature representation.
2) The cross-attention fusion module (CAFM) introduces the cross-attention mechanism [5] to establish semantic associations between different modalities. Subsequently, a channel embedding with a skip-connected structure is used to enhance feature representation capability.

The intuition that we provide is tested through an extensive ablation study of the design modules and parameters. Overall, our network outperforms state-of-the-art methods in a semantic segmentation task on two publicly available multimodal crop classification datasets. Our main contributions are summarized in the following.

1) We present a unified cross-modal fusion framework designed for crop classification across various modalities, employing a decoupled spatiotemporal encoder to separately model spatial and temporal features of SITS.
2) We rethink the fusion strategy between multimodal remote sensing images and prove that deep interaction within features is crucial for integrating complementary information between different modalities.
3) For the first time, we introduce the cross-attention mechanism in the crop classification task to facilitate semantic context exchange between multimodal SITS.

II. Related Work

A. Crop Classification for SITS

Crop classification is an important subscribe of land cover classification, which allocates specific crop categories to a set of required positions on a geospatial grid. To complete this task successfully, modeling the phenological cycles of crops is crucial [21]. Therefore, compared to single modal remote sensing images, the input of the model needs to be a satellite image series of length T. Initial studies often incorporate extra domain knowledge such as vegetation indices to reflect crop phenology and guide the process of feature extraction [21], [22]. Recently, deep neural networks (DNNs) trained on raw remote sensing images have demonstrated superior performance compared to these methods [23], [24]. At the object level (classification task), sequences of geospatial parcels are used as input of DNN, learning the probability sequence of K crop categories. Pelletier et al. [25] employ the 1-D temporal convolutional framework for SITS classification, while Cropformer [26] uses a Transformer [27] architecture. DuPLo [28] is composed of a set of CNN and RNN streams, designed to integrate CNN's spatial perception with RNN's temporal modeling. At the pixel level (semantic segmentation task), each pixel is assigned a semantic label, learning a map $f(\mathbf{X}) \in \mathbb{R}^{H\times W\times K}$. Rußwurm and Körner [29] explore the effectiveness of several typical convolutional RNN encoders (CLSTM and CGRU) [30] for SITS semantic segmentation tasks; it proves that these encoders can automatically extract the spatiotemporal features of the raw remote sensing image sequences, thereby achieving land cover classification. The choice between directly employing 3DCNN to extract spatiotemporal features and jointly utilizing 2-D CNN and convolutional RNN variants for spatialwise and temporalwise modeling is discussed in [31]. For a similar purpose,

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Martinez et al. [32] use a feature pyramid network (FPN) [33] architecture combined with a CLSTM temporal encoder. Tarasiou et al. [34] replace the decoder based on the UNET3D architecture, more suitable for contrastive learning. U-TAE [35] follows the 2-D UNet encoder architecture and introduces a weight-sharing temporal attention mechanism at each feature level to compress temporal features. On this basis, [36] extends it to a multimodal input architecture. TSVit [37] proposes a full-attention framework for SITS processing based on Vision Transformer (ViT) [38].

B. Multimodal Deep Learning in Remote Sensing

Previous works have achieved high performance on DNN based on single modal remote sensing images [39], [40]. However, in challenging real-world conditions, multimodal perception is required to achieve reliable and comprehensive scene understanding. The integration of various sensors, including optical, multispectral, and radar, has been extensively researched for their organic combination [41], [42].

Due to the heterogeneity of different sensor data, its focus is to design a framework that can effectively fuse different modalities of remote sensing images. For multimodal deep learning in remote sensing, there are three dominant strategies. The first mainstream paradigm involves integrating all acquisition channels of multimodal time series in the input stage of the network, thereby forming a cohesive whole comprising multimodal and multitemporal pixel features [23], [43]. Although these works have demonstrated the feasibility of learning multimodal remote sensing images within a single shared network, the training process may encounter information confusion due to the distinct feature expressions across different modalities. The second paradigm aims to develop two parallel modal branches and fuse them in the feature dimension. The dual-branch network architecture can better integrate the information of different modalities or subtasks. Wang et al. [44] propose a dual-branch multitask framework balance change detection and superpixels to achieve globally optimal detection performance. Ienco et al. [45] combine dedicated convolutional and recurrent-convolutional networks to encode optical and radar time series data separately, with all feature sequences connected and decoded by a multilayer perceptron (MLP). Tom et al. [46] train three encoders to transform three distinct multimodal acquisitions into a shared feature space for monitoring lake ice. The third paradigm involves fusing the semantic segmentation outputs from various modalities at the decision level to enhance the accuracy of the final semantic segmentation results. Le Bris et al. [47] propose a decision-level fusion scheme for fine-grained land cover classification, which fuses high-resolution optical images with low-resolution hyperspectral images.

While the deep learning framework with multimodal input has been widely used in remote sensing, previous works only concatenate tensors in channel dimension, lacking deeper interactions between different modalities. Joze et al. [48] utilize squeezing and excitation operations, utilizing knowledge from multiple modes to recalibrate channel features in each CNN stream. In a more global setting, we perform fusion at each level through cross-modal feature transfer and cross-attentional exchange to enhance dense crop semantic inference.

III. Proposed Method

A. Overview of the Network

To effectively fuse multimodal and multitemporal remote sensing image data for crop classification tasks, in this article, we present a two-stage fusion network called CMINet. The architecture of CMINet is shown in Fig. 2. The dual-branch network employs a decoupled spatially–temporally encoded process, utilizing the same structure, to fully exploit the information SITS from different modalities. Overall, we adopt an upsampling–downsampling architecture similar to U-Net [49] with four convolutional layers for SITS semantic segmentation. Image time series X^m of size $T \times C_m \times H \times W$ is input into each modality branch separately, where T represents the number of the length in the time dimension, C_m represents the number of channels from modality m, and $H \times W$ represents the spatial scale of the patch. At the lowest resolution of downsampling, a lightweight temporal attention encoder [50] is used to collapse the temporal dimension of each feature series, sharing weights at each resolution. In order to enhance the information transmission between modality branches and form complementary features, we resort to the interactive fusion strategy, as shown in Fig. 1(d). Specifically, a CMTM is designed to transfer knowledge from one modality branch to the other at each level of the spatial encoder. Thus, feature sequences obtain channelwise and spatialwise rectification before being sent to the next layer, softening the impact of noise from various modalities. Besides, a CAFM merges features compressed by temporal encoders belonging to the same level into one single feature map.

B. Spatial Encoder

We use a convolutional encoder called ε with four layers for space encoding. Each layer in the encoder consists of three main operations: convolution, rectified linear unit (ReLU) [51], and normalization. The encoder is applied to each modality and each timestamp of the input data. Starting from the second layer, each layer begins with a stride convolution. This operation applies a convolutional filter to the input feature map but with a specific stride value. This stride value determines the amount of downsampling performed on the feature map. As a result, the resolution of the feature image is divided by 2 compared to the previous layer.

For each modality branch m and each timestamp t, ε_l means the spatial encoder at layer l that takes the feature map from the previous layer $F_t^{m,l-1}$ as input and outputs a feature map $F_t^{m,l}$ of size $C_l \times H_l \times W_l$, where $H_l = (H_{l-1}/2)$ and $W_l = (W_{l-1}/2)$. The resulting feature sequences $F^{m,l}$ are then stacked in the time dimension to form a sequence of feature maps of size $T \times C_l \times H_l \times W_l$

$$F^{m,l} = \left[\varepsilon_l\left(F_t^{m,l-1}\right)\right]_{t=0}^{T} \tag{1}$$

with $e_t^{m,0}$ representing X^m and $[\cdot]$ serving as the concatenation operation along the time dimension.

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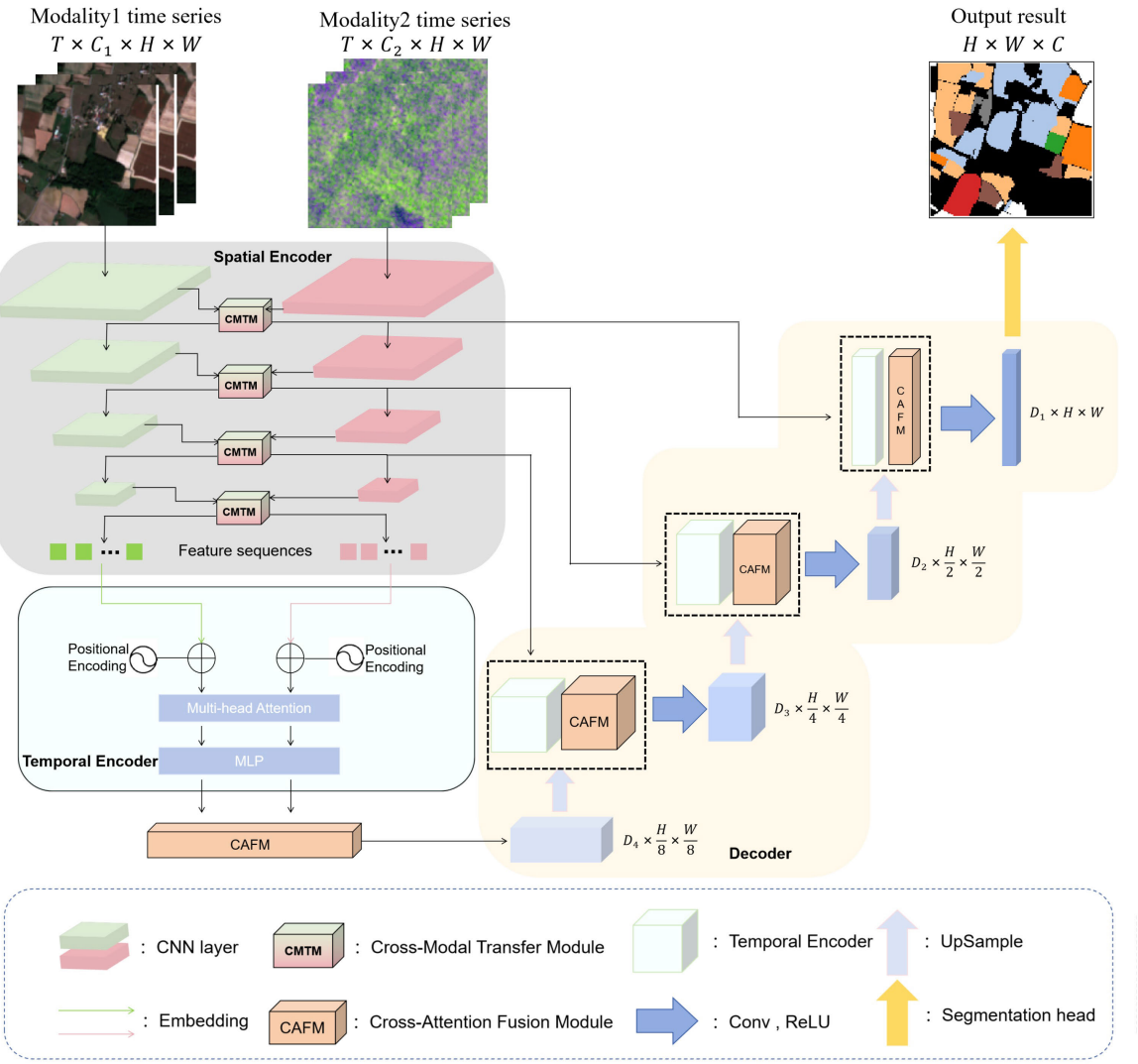


Fig. 2. Overview of CMINet for crop classification, consisting of the CNN-based spatial encoder and the attention-based temporal encoder on the left and the decoder shown on the right. Four CMTM modules are used to perform knowledge transfer between the two modality branches at each resolution layer. The CAFM introduces a cross-attention mechanism [5] to merge the time-encoder compressed features before feeding them to the decoder.

C. Cross-Modal Transfer Module

As above mentioned, remote sensing images from different sensors are usually complementary [13], [14]. However, they are often affected by noise measurements. The noise information of one modality can be filtered and rectified by receiving information from another modality. For example, cloud cover is one of the most common types of noise in optical remote sensing images. Radar images are frequently employed as auxiliary data to mitigate cloud cover in optical imagery due to their superior penetrability and immunity to cloud obstructions [52]. For this purpose, we propose a novel CMTM in Fig. 3 to perform knowledge transmission between modality branches at each layer of the spatial encoder. To weaken the effects of noise measurements and uncertainties from different modalities, CMTM processes channelwise and spatialwise transfer, offering an all-round rectification.

1) Channelwise Transfer: At the layer l of the spatial encoder, feature sequences from two modality branches $F^{m_1,l}, F^{m_2,l} \in \mathbb{R}^{T\times C_l\times H_l\times W_l}$ are embedded along the spatial axis into two weight masks $W^C_{m_1,l}, W^C_{m_2,l} \in \mathbb{R}^{T\times C_l}$. Previous methods [48], [53] consider only average pooling to capture global information from feature sequences, leading to information loss. We utilize both global max pooling and global average pooling to $F^{m_1,l}$ and $F^{m_2,l}$ in channel dimension to preserve more information. Then, we concatenate the four pooled tensors along the channel axis, getting $Y \in \mathbb{R}^{T\times 4C_l}$. An MLP is applied after that, followed by a sigmoid function to standardize a reasonable range of weights from Y, which will be divided into $W^C_{m_1,l}$ and $W^C_{m_2,l}$

$$Y = \text{concat}\big[\text{Avgpool}\big(F^{m_1,l}\big), \text{Maxpool}\big(F^{m_1,l}\big), \text{Avgpool}\big(F^{m_2,l}\big), \text{Maxpool}\big(F^{m_2,l}\big), \text{dim} = 1\big] \quad (2)$$

$$W^C_{m_1,l}, W^C_{m_2,l} = \mathcal{F}_{\text{split}}\left(\text{sigmoid}\big(\mathcal{F}_{mlp}(\mathbf{Y})\big)\right). \quad (3)$$

2) Spatialwise Transfer: We archive spatialwise transfer by utilizing the spatial relationships between features. Unlike channelwise transfer that focuses on transmitting global information, spatialwise transfer focuses on "where" is an

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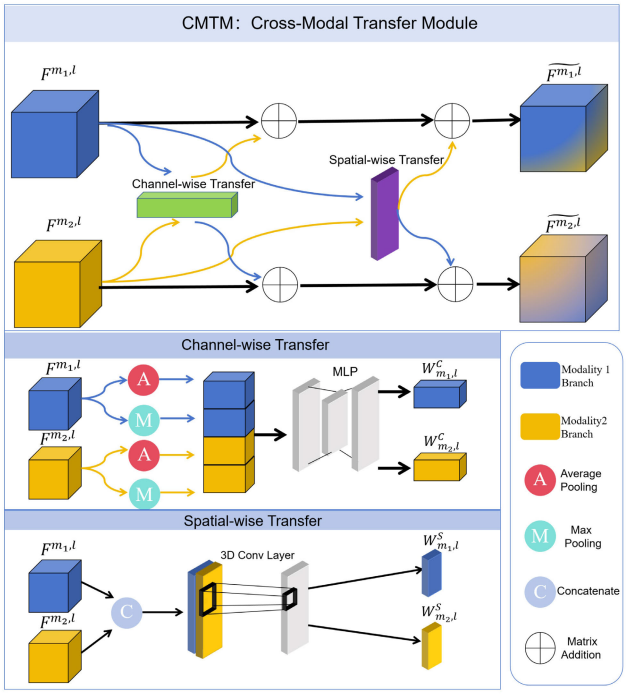


Fig. 3. Architecture of CMTM for two modality branches. Feature sequences from different modality branches are rectified by facilitating channelwise and spatialwise transfer.

information-rich part to calibrate features locally, which is a supplement to channelwise transfer. The dual-branch inputs $F^{m_1,l}$ and $F^{m_2,l}$ at the l_{th} layer of the spatial encoder are concatenated in channel dimension and embedded into two spatial weight masks $W^S_{m_1,l}, W^S_{m_2,l} \in \mathbb{R}^{T\times H\times W}$. The embedding operation consists of two $1\times 1\times 1$ 3-D convolution layers joined by an ReLU. Similar to channelwise transfer, a sigmoid function is used to obtain the feature sequences $F \in \mathbb{R}^{T\times H\times W\times 2}$, which is subsequently spilt into two weight masks

$$\mathbf{F} = \text{Conv}_{1\times1\times1}\left(\text{ReLU}\left(\text{Conv}_{1\times1\times1}\left[F^{m_1,l}, F^{m_2,l}\right]\right)\right) \tag{4}$$

$$W^S_{m_1,l}, W^S_{m_2,l} = \mathcal{F}_{\text{split}}\left(\text{sigmoid}(\mathbf{F})\right) \tag{5}$$

where $[\cdot]$ represents the concatenation along the channel dimension.

The learned weight masks of each modality are multiplied by the features of each modality, and the results are transmitted as supplementary knowledge to another modality. The whole transfer process is formulated as

$$\widetilde{F^{m_1,l}} = F^{m_1,l} + \lambda_c\left(W^C_{m_2,l}\odot F^{m_2,l}\right) + \lambda_s\left(W^S_{m_2,l} * F^{m_2,l}\right) \tag{6}$$

$$\widetilde{F^{m_2,l}} = F^{m_2,l} + \lambda_c\left(W^C_{m_1,l}\odot F^{m_1,l}\right) + \lambda_s\left(W^S_{m_1,l} * F^{m_1,l}\right) \tag{7}$$

where $\odot$ means the channelwise multiplication and $*$ means the spatialwise multiplication. Both λ_c and λ_s are set up as 0.5 as default. Rectified feature sequences $\widetilde{F^{m_1,l}}$ and $\widetilde{F^{m_2,l}}$ will be inputted into the next layer of the spatial encoder.

D. Temporal Encoder

Redundant information is difficult to handle within an entangled spatiotemporal space [54]. Therefore, our CMINet employs a decoupled spatial and temporal encoding framework, modeling the spatial and temporal representations of SITS separately. After completing the cross-modal transfer process, a temporal encoder is applied to compress each feature sequence from both modality branches in the time dimension, enabling their use as skip connections. Taking into account both effectiveness and processing efficiency, we opt for the lightweight temporal attention encoder [50] as our temporal encoder, which is a simplified version of the multihead self-attention network [27]. As illustrated in Fig. 2, the outputs of multihead attention are concatenated and processed by an MLP to obtain the desired size.

For each modality m, we apply our temporal encoder φ with a G header to each pixel of a sequence of rectified features of minimum resolution L, thus generating G temporal attention masks for each pixel. These masks can be organized into a G tensor with values between 0 and 1 in the shape of $T \times H_L. \times H_L.$.

For each modality m, we utilize our temporal encoder φ with P heads on each pixel of the rectified feature sequences at the lowest resolution L, thereby generating P temporal attention masks for each pixel. These masks can be organized into P tensors with values ranging from 0 to 1 and a shape of $T \times H_L \times H_L$

$$a^{m,L,1}, \ldots, a^{m,L,P} = \varphi\left(\widetilde{F^{m,L}}\right). \tag{8}$$

To extend these attention masks to all resolution layers l, we use bilinear interpolation to interpolate and obtain masks of each modality $a^{m,l,g}$ of size $T\times H_l\times H_l$. Subsequently, rectified feature sequences from each modality $\widetilde{F^{m,l}}$ are divided into P consecutive groups along the channel axis. In group p, the feature sequence $\widetilde{F^{m,l,p}}$ is averaged in the time dimension utilizing $a^{m,l,p}$ as weights. Results are concatenated along channel dimension and processed by a convolutional layer of size 1×1. Thus, we obtain feature maps of each layer $f^{m,l}$ of size $D_l \times H_l \times W_l$

$$f^{m,l} = \text{Conv}_{1\times1}\left(\left[\sum_{t=1}^{T} a_t^{m,l,p}\otimes \widetilde{F_t^{m,l,p}}\right]_{p=1}^{P}\right) \tag{9}$$

where $[\cdot]$ serves as the concatenation operation along the channel dimension and $\otimes$ represents the elemental multiplication of the channel broadcast.

E. Cross-Attention Fusion Module

After extracting feature maps of each layer $f^{m,l}$, we design a CAFM as illustrated in Fig. 4 to combine features from two modality branches. Within this module, we introduce a cross-attention mechanism [5] to achieve the global context representations of each modality and exchange them. Afterward, the concatenated features were transformed back to their original size through a channel embedding process.

The cross-attention mechanism for exchanging information across modalities in this module is based on the traditional

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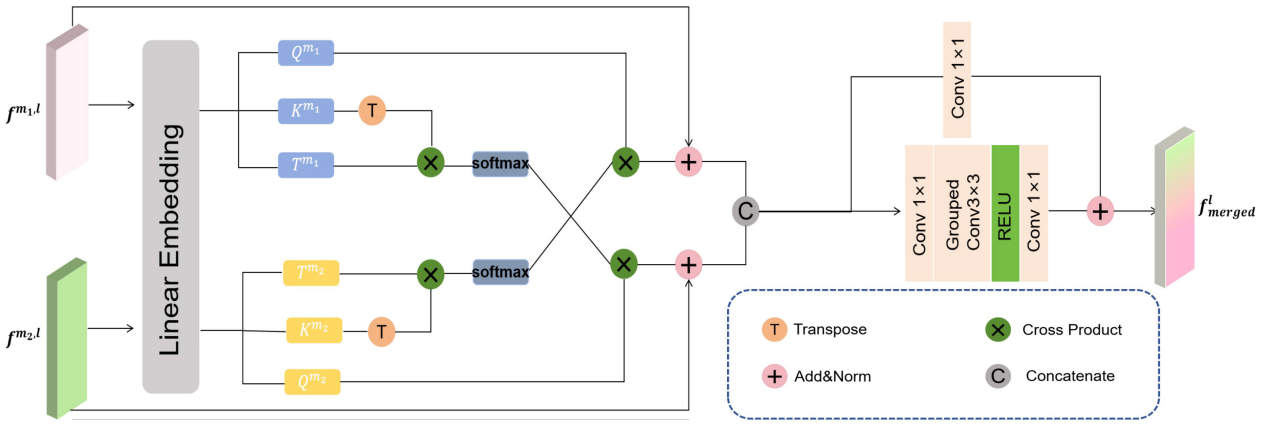


Fig. 4. Flowchart of the introduced CAFM. This module includes a cross-attention stage and a channel embedding stage, which realizes context exchange and fusion of the features from two modality branches.

self-attention [27]. Initially, we flatten our input feature maps $f^{m,l} \in \mathbb{R}^{D_l \times H_l \times W_l}$ into vectors of size $D_l \times N_l$, where $N_l = H_l \times W_l$. Subsequently, a linear embedding operation is applied to generate Query ($\mathbf{Q}$), Key ($\mathbf{K}$), and Value ($\mathbf{V}$) like the traditional self-attention mechanism. $\mathbf{Q}$ is the dot product with the transpose of $\mathbf{K}$ to generate an intermediate matrix for measuring attention scores, which causes a high memory usage. Instead, efficient attention [55] calculates a global vector $\mathbf{G} = \mathbf{K}^{\mathrm{T}}\mathbf{V}$ with substantially less memory and computational costs. Following this way, we calculate the global vectors $\mathbf{G}^l_{m_1}$ and $\mathbf{G}^l_{m_2}$ from both modality branches separately. The global vectors are cross-product with query vector $\mathbf{Q}$ from the other modality branch to complete the cross-attention operation. This operation facilitates contextual interaction from a sequence perspective and serves as an effective complement to the overall information flow within the CMTM module. The cross-attention process is formulated as

$$\mathbf{U}^l_{m_1} = \mathbf{Q}^l_{m_1}\,\mathrm{SoftMax}\left(\mathbf{G}^l_{m_2}\right) \tag{10}$$

$$\mathbf{U}^l_{m_2} = \mathbf{Q}^l_{m_2}\,\mathrm{SoftMax}\left(\mathbf{G}^l_{m_1}\right) \tag{11}$$

where $\mathbf{U}^l_{m_1}$ and $\mathbf{U}^l_{m_2}$ mean the cross-attention result of each modality branch at the l_{th} resolution layer. They are concatenated as $\mathbf{U}$ and reshaped to size of $2D_l \times H_l \times W_l$. To restore the concatenated feature to its original size, a channel embedding is adopted through a 1×1 convolution layer. Besides, in the process of channelwise fusion, incorporating the information of surrounding pixels can also help with crop classification tasks. Therefore, we add a 3×3 grouped convolution layer as an extra stream to build a skip-connection structure. Thus, the concatenated features are fused into the final feature map $f^l_{\mathrm{merged}} \in \mathbb{R}^{D_l \times H_l \times W_l}$.

The merged feature map from the lowest resolution layer serves as the input to the decoder, which comprises several upsampling convolutional blocks. In contrast, feature maps from other layers are employed as skip connections. During the decoding process, the decoder blocks restore the spatial resolution to $H \times W$ by concatenating the skip connections from corresponding layers. Ultimately, a segmentation head generates the final crop classification results.

IV. Experiments

A. Datasets

We use two publicly available multimodal SITS datasets for crop classification, the PASITS-R dataset [36] and the South Sudan dataset [31], to verify our proposed CMINet.

1) PASITS-R Dataset [36]: The PASITS-R dataset is the multimodal version of the PASITS dataset [35]. It comprises 2433 SITS samples of 128 × 128 pixels, including ten spectral bands from Sentinel-2 (excluding atmospheric bands B01, B09, and B10) and three radar bands (VV polarization, VH polarization, and VV/VH values) from Sentinel-1. These samples are drawn from four distinct regions of France, characterized by diverse climates and crop distributions, encompassing 18 different crop types. Both image modalities possess a resolution of 10 m, and all data are sourced from the publicly accessible Sentinel satellite platform. For each sample, the PASITS-R dataset compiles a comprehensive collection of satellite images, spanning from September 2018 to November 2019, thereby constructing the final SITS.

2) South Sudan Dataset [31]: The South Sudan dataset is made up of 687 growing season SITS samples of crop fields from three satellites (Sentinel-1, Sentinel-2, and PlanetScope) in 2017 in South Sudan. Among them, the size of the Sentinel image is 64 × 64 pixels, while the size of the PlanetScope image is 128 × 128 pixels. Samples are labeled for four crop types: maize, groundnut, rice, and sorghum. Each sample includes three Sentinel-1 bands (VV, VH, and VH/VV), ten Sentinel-2 bands (red, green, blue, near-infrared, four red-edge bands, and two short wave infrared bands), and four PlanetScope bands (red, green, blue, and near-infrared), varying from 25 to 100+ observations.

B. Implementation Details

In all the experiments presented in this article, we have used the data splits provided in the relevant publications and performed an equal amount of training epochs to ensure fairness in the comparisons. For the PASITS-R dataset, the raw 128 × 128 size multispectral and radar image series are used as inputs for the two modality branches. For the

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TABLE I
CROP CLASSIFICATION ACCURACY OF DIFFERENT APPROACHES ON THE PASITS-R DATASET AND THE SOUTH SUDAN DATASET

Method	Multimodal	Parameter	PASITS-R			South Sudan		
		(M)	OA(%)	mF1(%)	mIoU(%)	OA(%)	mF1(%)	mIoU(%)
BiConvGRU[29]	N	4.5	80.5	69.3	56.2	86.4	64.9	55.5
FPN-ConvLSTM[32]	N	1.2	81.9	71.0	59.5	85.1	64.8	54.8
TSViT[37]	N	1.7	83.4	77.9	65.4	87.6	68.9	60.3
UNET3Df[34]	Y	7.2	82.5	73.8	60.6	87.3	65.6	56.9
UNET3D[31]	Y	6.2	82.7	74.6	61.8	88.1	67.6	59.5
U-ConvLSTM[31]	Y	2.6	83.0	75.5	62.7	86.7	65.3	55.7
U-TAE[36]	Y	1.7	84.2	78.5	66.3	89.8	71.0	62.6
CMINet(ours)	Y	2.1	**85.3**	**80.2**	**68.3**	**92.4**	**72.1**	**65.4**

South Sudan dataset, the Sentinel image series and PlanetScope image series are used as inputs for the two modality branches, respectively. The PlanetScope images are resized to 64 × 64 to match the size of the Sentinel images. All sequences in both datasets are padded to 128 in the time dimension to ensure consistency in the size of the input tensors. For training our CMINet, we use the Adam optimizer [56] with a leering rate of 0.001. All networks are trained on a single NVIDIA A30 GPU with 24 GB of VRAM for 100 epochs in batches of four time series.

1) Loss Function: All models are trained with a masked cross-entropy loss in order to exclude the effect of the background class.

2) Evaluation Metrics: The accuracy of crop classification is evaluated using three metrics: overall accuracy (OA), mean $F1$ score (mF1), and mean intersection over union (mIoU). The formulas for calculating OA, mF1, and mIoU are given as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{12}$$

$$\mathrm{mF1} = \frac{1}{n+1} \sum_{i=0}^{n} \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{13}$$

$$\mathrm{mIoU} = \frac{1}{n+1} \sum_{i=0}^{n} \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{14}$$

where TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative, respectively, and n represents the number of categories.

C. Comparison With SOTA

We compare the performance of our proposed CMINet on the PASITS-R dataset and the South Sudan dataset with seven representative state-of-the-art methods for crop classification from SITS, namely, BiConvGRU [29], FPN-ConvLSTM [32], TSViT [37], UNET3Df [34], UNET3D [31], U-ConvLSTM [31], and U-TAE [36]. Among these methods, BiConvGRU, FPN-ConvLSTM, and TSViT only consider information from the primary modality, i.e., the optical image series. By contrast, other approaches take into account cross-modal data. Comparing advanced single-modal methods with those that incorporate multimodal input can reveal the benefits of leveraging complementary information from different modalities, such as radar images, in crop classification tasks. The quantitative results are listed in Table I. We can find out that our CMINet achieves a leading performance in the three evaluation metrics of OA, mF1, and mIoU compared to existing methods on both datasets. In addition, Table I also shows the parameters of each method to reflect the complexity of the model and the usage of computational resources. Compared to methods such as UNET3Df and UNET3D that employ 3-D convolution to uniformly extract spatiotemporal features, we design a decoupled framework modeling spatial and temporal representations from SITS separately, greatly reducing the number of model parameters. In comparison to lightweight methods, CMINet improves the accuracy of crop classification with minimal parameter increase.

1) Performance Comparison on the PASITS-R Dataset: As illustrated in Table I, CMINet has achieved the highest OA (85.3%), mF1 (80.2%), and mIoU (68.3%) on the PASITS-R dataset. Compared to the novel single-modal method TsViT, our CMINet provides a substantial improvement of 1.9% on OA, 2.3% on mF1, and 2.9% on mIoU. This demonstrates that CMINet can effectively integrate complementary information from radar image time series, providing more comprehensive representations of crop parcels. Among all multimodal input methods, CMINet also performs the best, exceeding the second-best U-TAE by OA of 1.1%, mF1 of 1.7%, and mIoU of 2.0%. This can be explained by the fact that our proposed cross-modal transfer and introduced cross-attention fusion strategy successfully enhance the information interaction between different modal branches. Fig. 5 shows four visualization examples of the crop classification obtained by all mentioned methods from top to bottom. From the Sentinel-2 images in Fig. 5(a1)–(d1), we can see that the arrangement of crop fields is intricate and dense, and it is difficult to distinguish specific crop types. The temporal attention encoder in CMINet can automatically evaluate the importance of each phase in SITS during the training process, effectively capturing the periodic changes in optical and polarization reflections of various crop plots, constructing phenological cycles of various crops, achieving accurate identification of crops in Fig. 5(a12)–(d12). Furthermore, the cross-attention mechanism in CAFM conducts sufficient semantic context exchange before merging and introduces an extra stream during the fusion stage to increase perception of surrounding pixels,

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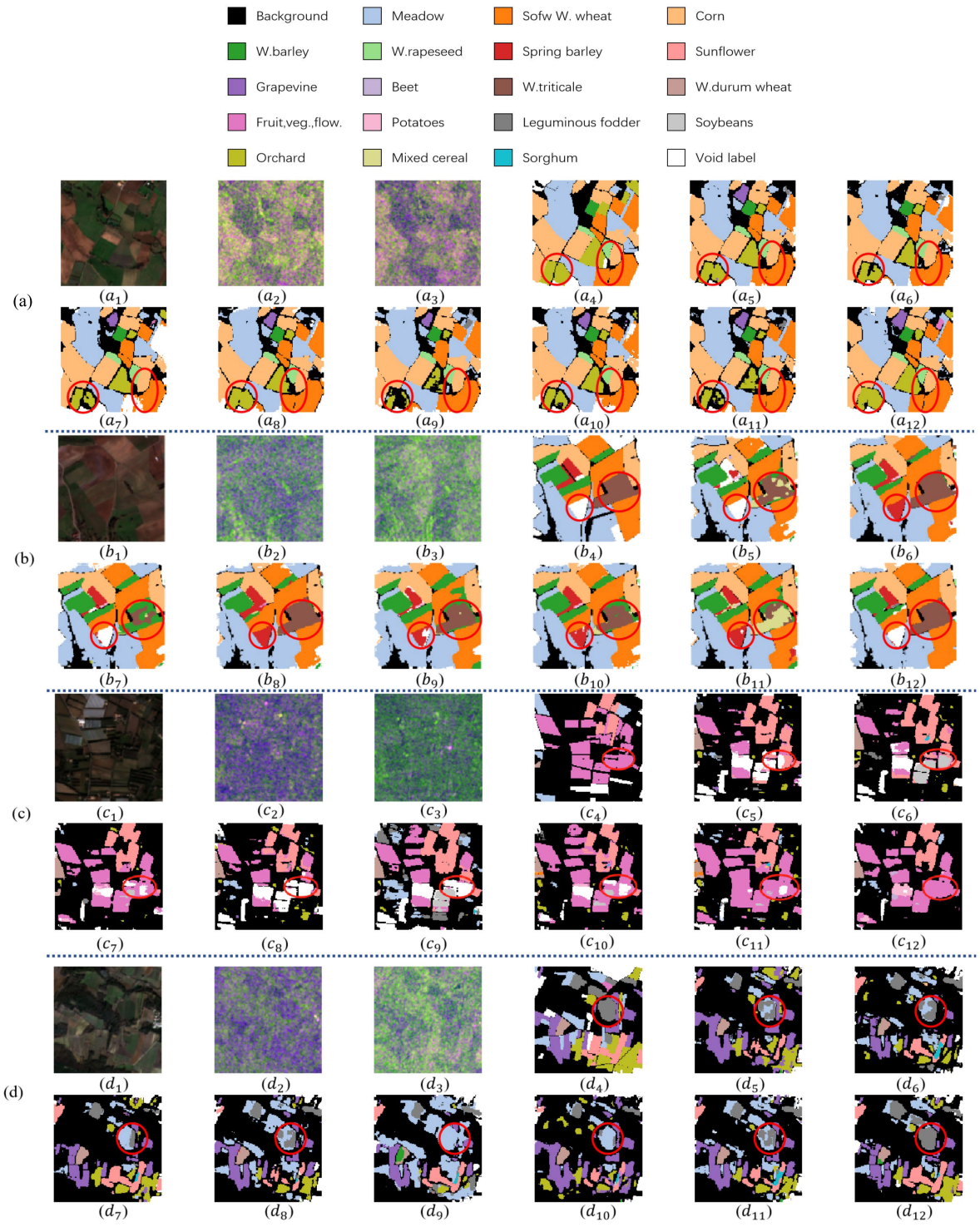


Fig. 5. Four examples of crop classification results on the PASITS-R dataset. (a1)–(d1) Sentinel-2 images. (a2)–(d2) Sentinel-1A images. (a3)–(d3) Sentinel-1D images. (a4)–(d4) Annotations. (a5)–(d5) BiconvGRU classification results. (a6)–(d6) FPN-ConvLSTM classification results. (a7)–(d7) TSViT classification results. (a8)–(d8) UNET3Df classification results. (a9)–(d9) UNET3D classification results. (a10)–(d10) U-ConvLSTM classification results. (a11)–(d11) U-TAE classification results. (a12)–(d12) CMINet classification results.

which helps identify a complete crop field object and edge detection. There are several red circles in Fig. 5(a12)–(d12), we can find that the existing methods may lead to misclassification and omission for small farmlands, and one complete crop field object may be filled with multiple crop pixels. In contrast, CMINet is capable of accurately classifying complete fields irrespective of their size while maintaining clearer field edges.

2) Performance Comparison on the South Sudan Dataset: The experiments on the South Sudan dataset obtain similar results to those on the PASITS-R dataset shown in Table I. The corresponding OA, mF1, and mIoU values are 92.4%, 72.1%,

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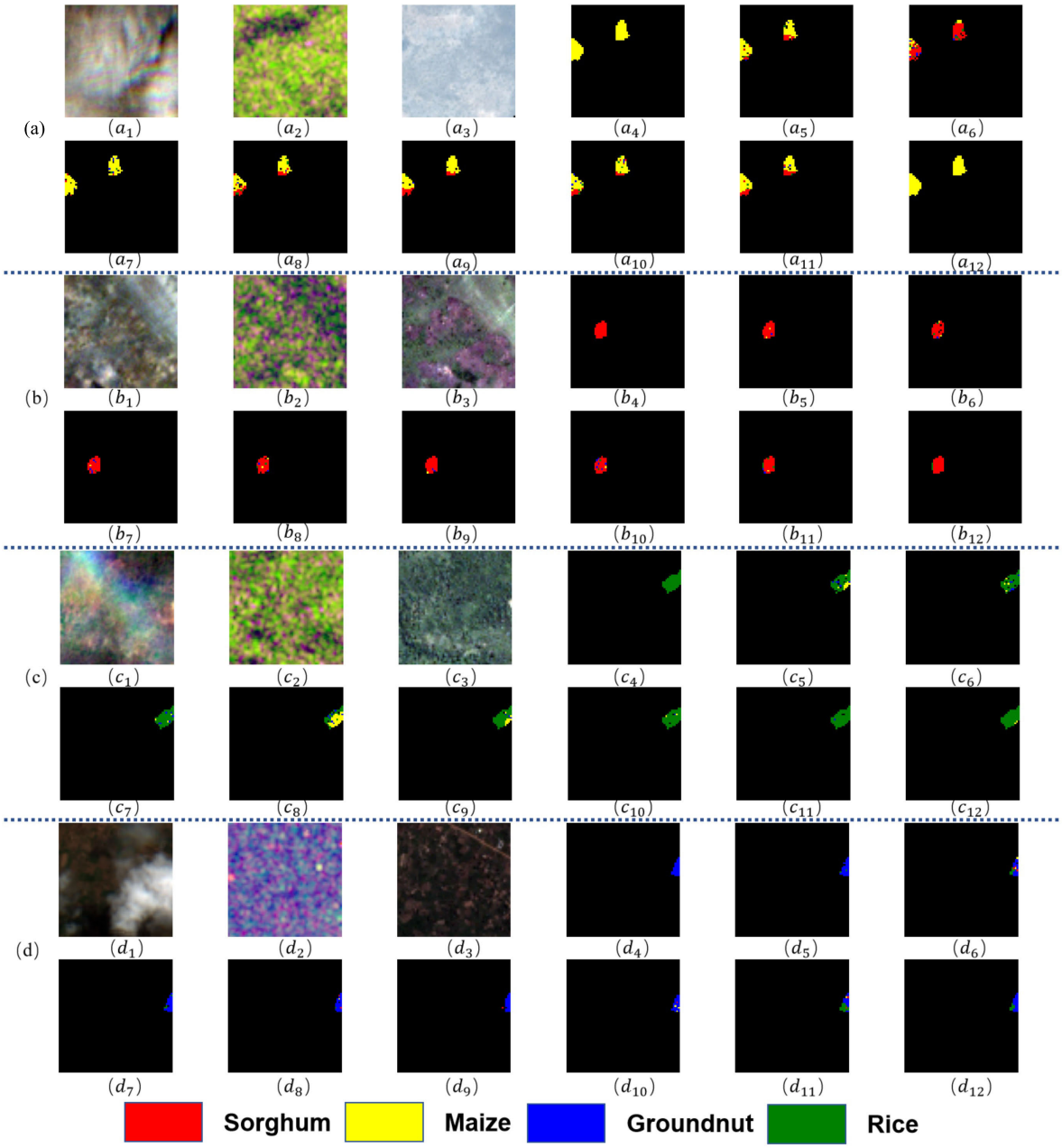


Fig. 6. Four examples of crop classification results on the South Sudan dataset. (a1)–(d1) Sentinel-2 images. (a2)–(d2) Sentinel-1A images. (a3)–(d3) Sentinel-1D images. (a4)–(d4) Annotations. (a5)–(d5) BiconvGRU classification results. (a6)–(d6) FPN-ConvLSTM classification results. (a7)–(d7) TSViT classification results. (a8)–(d8) UNET3Df classification results. (a9)–(d9) UNET3D classification results. (a10)–(d10) U-ConvLSTM classification results. (a11)–(d11) U-TAE classification results. (a12)–(d1) CMINet classification results.

and 65.4%, respectively, which corresponds to increases of 2.6%, 1.1%, and 2.8%, respectively, over U-TAE. Fig. 6 shows four visualization examples of the crop classification results from the South Sudan dataset for all mentioned methods. We can see that the distribution of fields in South Sudan is very sparse and scattered, and optical images are greatly affected by cloud cover. Considering the strong penetrability of radar image series, our proposed CMTM in CMINet performs knowledge transfer across modality branches, mitigating the effects of uncertainties and noise measurements from different modalities. It can be observed that CMINet can effectively eliminate interference from cloud cover and achieve more accurate identification of crop fields in South Sudan compared to existing methods from Fig. 6(a4)–(d12).

D. Ablation Study

We conducted a series of ablation studies, systematically removing individual modules while maintaining the dual-branch architecture, to validate the effectiveness of each proposed component within CMINet. To facilitate knowledge transmission and information exchange between two modality branches, we have implemented two modules: CMTM and CAFM. As shown in Table II, we take these two modules out

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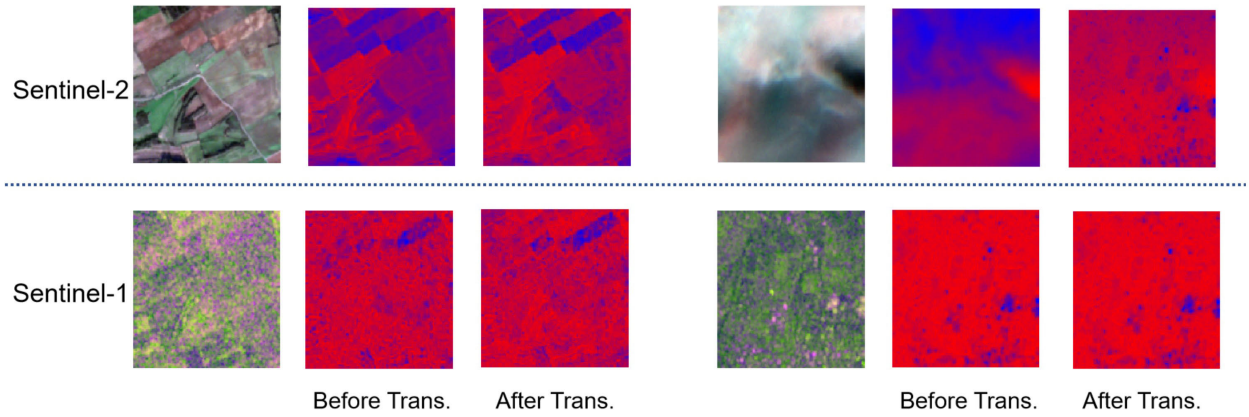


Fig. 7. Visualization of a certain phase in the feature sequences at the first layer of the spatial encoder and the rectified features through CMTM knowledge transfer.

TABLE II

ABLATION STUDY OF THE PROPOSED METHODS ON THE PASITS-R DATASET, WHERE CMTM REFERS TO THE CROSS-MODAL TRANSFER MODULE AND CAFM REFERS TO THE CROSS-ATTENTION FUSION MODULE. THE RESULTS SHOW THE EFFECTIVENESS OF CROSS-MODAL KNOWLEDGE TRANSFER AND INFORMATION EXCHANGE

CMTM	CAFM	OA(%)	mF1(%)	mIoU(%)
×	×	84.1	76.9	64.4
✓	×	84.6	78.6	66.7
×	✓	84.7	78.1	66.3
✓	✓	**85.3**	**80.2**	**68.3**

TABLE III

CROP CLASSIFICATION ACCURACY OF THE PASITS-R DATASET OBTAINED BY CMINET WITH DIFFERENT INPUTS

Input	OA(%)	mF1(%)	mIoU(%)
Single date Sentinel-2(September)	75.4	49.8	26.7
Single modal time series(Sentinel-2)	83.2	75.9	63.1
Sentinel-1+Sentinel-2 time series	**85.3**	**80.2**	**68.3**

of the network separately. When CMTM is removed, the two modality branches in the spatial encoder will independently extract their respective features without any interaction. When CAFM is removed, the two features compressed by the time encoder at each resolution level will be fused by taking the average in the channel dimension.

The findings in Table II suggest that both CMTM and CAFM are essential for the proposed CMINet to effectively integrate cross-modal information for accurate crop classification. Compared to the baseline, the use of CMTM alone results in an increase of 0.5% in OA, 1.7% in mF1, and 2.3% in mIoU, the use of CAFM alone results in an increase of 0.6% in OA, 1.2% in mF1, and 1.9% in mIoU, and together CMTM and CAFM result in an increase of 1.2% in OA, 3.3% in mF1, and 3.9% in mIoU. More specifically, CMTM rectifies features by transmitting information from another modality branch, thereby enhancing the model's robustness under varying conditions. By introducing a cross-attention mechanism, CAFM effectively integrates the semantic context information between different modalities, allowing the model to focus on the key information of each modality to better understand and process more complex scenarios.

To effectively illustrate the critical role of cross-modal knowledge transfer, we visualize a specific phase in the feature sequence before and after CMTM rectification in Fig. 7. Generally, following cross-modal knowledge transfer, both branches exhibit enhanced features. Specifically, the optical branch of Sentinel-2 enhances its capability to detect texture features of crop fields, whereas the radar branch of Sentinel-1 improves its ability to extract the boundaries and continuity of cultivated land. Moreover, when optical imagery is significantly obscured by clouds, the features are effectively corrected through the receipt of information from the radar mode.

V. DISCUSSION

A. Advantages of Multimodal and Multitemporal Input

1) Advantages of Multimodal Input: To further investigate the influence of multimodal input methods on crop classification tasks, we compare networks trained on single-modality image sequences from Sentinel-2 with networks trained on multimodal image series from both satellites. Based on Table III, the network trained on the multitemporal Sentinel-1 and Sentinel-2 image sequences demonstrates OA, mF1, and mIoU values of 85.3%, 80.2%, and 68.3%, respectively. These values are higher by 2.1, 4.3, and 5.2 percentage points compared to the network trained solely on Sentinel-2 image time series. It signifies that combining multimodal data allows deep networks to acquire more comprehensive and diverse information, thereby enhancing the accuracy and dependability of the deep networks. The information derived from Sentinel-2 multispectral images is abundant and intuitive, making it more advantageous for crop classification tasks. On the other hand, Sentinel-1 radar images possess the ability to penetrate clouds, rain, and snow weather and provide scattering characteristics of ground objects. These unique features serve as beneficial supplements to Sentinel-2 multispectral images, ultimately leading to an overall improvement in network performance.

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TABLE IV
CROP CLASSIFICATION ACCURACY OF THE PASITS-R DATASET OBTAINED BY CMINET WITH DIFFERENT FUSION STRATEGIES

Fusion strategy	OA(%)	mF1(%)	mIoU(%)
Input fusion	83.6	76.4	63.8
Feature fusion	84.1	76.9	64.4
Decision fusion	83.3	76.1	63.5
Interactive fusion	**85.3**	**80.2**	**68.3**

2) Advantages of Multitemporal Input: Furthermore, to demonstrate the efficacy of SITS in uncovering the phenological cycle of crops and improving classification accuracy, we utilize cloud-free single-temporal Sentinel-2 images from September for network training. The trained network is compared to a network that consumes SITS. As shown in Table III, when using single-phase Sentinel-2 imagery, there is a significant decline in the OA, mF1, and mIoU values of the trained network compared to using multiphase SITS. This indicates the necessity and importance of using SITS as input for crop classification tasks in order to explore the phenological patterns of crop growth. In practical applications, crop growth and development are influenced by various factors such as climatic conditions, human activities, and natural disasters. These factors result in different growth states and characteristics of crops in different periods. Single-phase imagery provides limited information and cannot fully capture the growth state and characteristics of crops. On the other hand, using SITS allows for complementarity between images from different periods. This enables a better exploration of the growth cycle and characteristics of crops, leading to more comprehensive and accurate crop information.

B. Discussion of Different Fusion Strategies

To discuss the superiority of our proposed interactive fusion strategy, we conducted experiments using the four different fusion strategies illustrated in Fig. 1. Input fusion involves concatenating the SITS of different modalities along the channel dimension at the input stage, encoding them jointly spatially and temporally. Feature fusion entails extracting features from the two modal branches, which are then averaged after compression by a time encoder. Decision fusion consists of employing two entirely independent networks to extract features from the two modalities and average the outcomes. Interactive fusion involves the use of our designed CAFM to fully blend the features from two modality branches. Moreover, our proposed CMTM performs cross-modal knowledge transfer at each layer. Experiments with different fusion methods are conducted according to the general framework of CMINet on the PASITS-R dataset.

As shown in Table IV, interactive fusion achieves the best performance with OA, mF1, and mIoU values, surpassing the second-best fusion strategies by 1.2%, 3.3%, and 3.9%, respectively. This suggests that interactive fusion can more effectively integrate multimodal information and generate more robust features for crop classification compared to current common fusion strategies. More specifically, whether the fusion strategy is input fusion, feature fusion, or decision fusion, it only involves a concatenation or averaging operation in the channel dimension, without considering the complementarity and balance between modalities, leading to an underestimation or overestimation of the impact of certain modalities on the final outcome. In addition, [57] and [58] use a multiloss to constrain the differences between classification results and features from different modalities or the difference between modalities. Compared with the feature interaction on the network side, this loss-level fusion relies more on the complexity of loss function design, which brings higher difficulty in parameter adjustment and cannot capture the nonlinear relationship between modalities.

C. Discussion of Uncertainty Brought by Different Environmental Conditions

Different geographic and natural environmental conditions introduce a certain amount of uncertainty into the classification model. In this article, we demonstrate from both the mechanism and data levels that our proposed CMINet can overcome the uncertainty brought to the network by the geographical and natural environments. From the mechanism level, there are two aspects in CMINet that can reduce the impact of geographical and natural environments on the network: 1) the temporal attention encoder assigns a lower weight to the phases that are affected by the geographical and natural environments, reducing their uncertainty in the classification results and 2) information interaction and knowledge transfer between different modalities can increase the robustness of the model and make it adaptable to more complex geographical and natural environments. For example, radar signals can penetrate tree canopies and vegetation to obtain information about ground objects. In a forest environment, the interaction between radar images and optical images can increase the network's ability to recognize crops in that environment.

From the data level, we have conducted experiments on two datasets with quite different environments. The PASITS-R dataset involves dense farmland environments in Europe, while the South Sudan dataset involves scattered farmland environments in Africa. There are considerable differences between the two in terms of climate and terrain. Our proposed CMINet has achieved good performance on both datasets, which, to some extent, shows that CMINet can eliminate the uncertainty brought to the network by the geographical and natural environments.

VI. CONCLUSION

In this work, a unified cross-modal integration network named CMINet is designed to handle multimodal SITS input and aims to explore a more effective fusion strategy of remote sensing data from different modalities for crop classification. In particular, a decoupled CNN-based framework equipped with proposed CMTM and CAFM is designed to perform spatial and temporal modeling of multimodal SITS separately. CMTM conducts knowledge transfer in both channel and spatial dimensions, presenting comprehensive feature rectification. CAFM introduces a cross-attention mechanism to

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enhance semantic context interaction and balance the integration of bimodal features. Comprehensive results from two datasets, PASTIS-R and South Sudan, prove that our proposed CMINet outperforms state-of-the-art crop classification approaches.

This study may face challenges in several aspects and deserves further exploration. First, more data modalities should be discussed. SRTM digital elevation data and nighttime light remote sensing images may be used as additional data modalities in future work. Second, this article does not specifically address the class imbalance phenomenon, which is prevalent in the field of crop classification. Introducing additional tasks into the existing architecture to constrain this phenomenon may be a good extension. Third, expanding the research to assimilate highly generalized domain knowledge, such as remote sensing indices, would be an interesting avenue for further investigation.

CRediT Authorship Contribution Statement

Yutong Hu: conceptualization, methodology, writing—original draft, validation, and visualization. **Qingwu Hu:** investigation and project administration. **Jiayuan Li:** investigation and supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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